Quantifying Uncertainty and Biases in Machine Learning

Week 5 - Lecture 4

Motivation

If we want to use a model to make predictions, it's useful to know **how certain we can be** in those predictions.

"All models are wrong but some are useful"

- **George E.P. Box**

The less certain we are in our predictions, the less useful our models are.

Motivation

- Having certainty in predictions is especially important in domains like healthcare, finance, and autonomous systems.

- Understanding uncertainty improves trust and decision-making.

How do we Quantify Uncertainty?

Commonly we use **Confidence Intervals (CIs).**

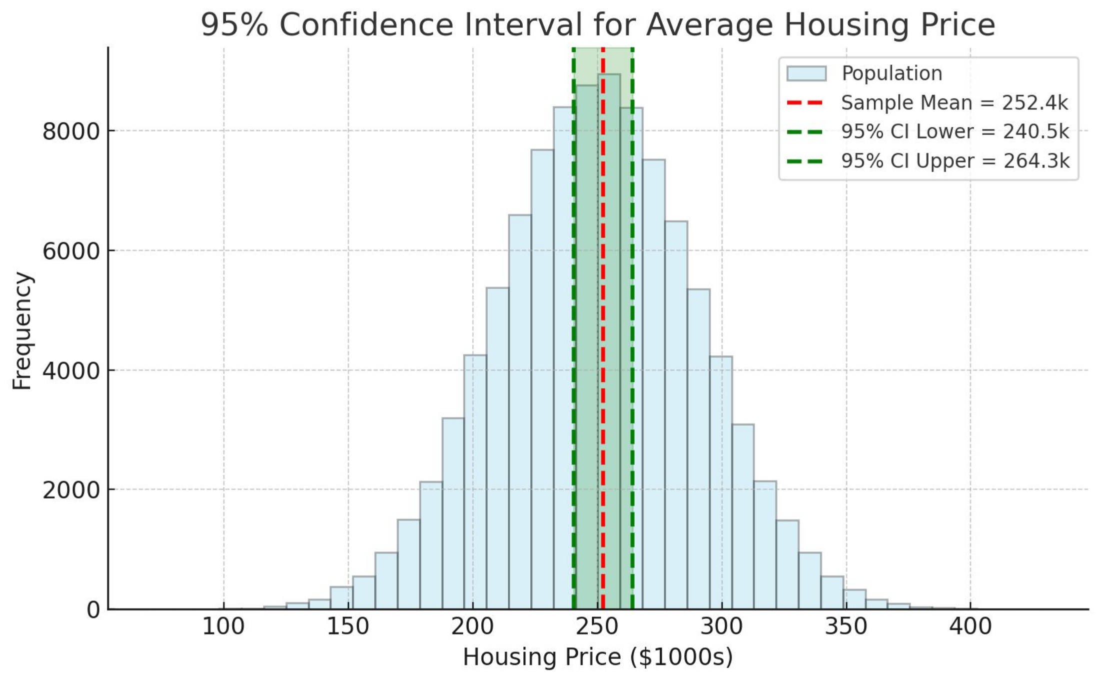
A **Confidence Interval** gives a **range of plausible values** for an **estimate**.

- e.g. "The model predicts the house price to be $250k, with a 95% CI of [$230k, $270k]."
- In other words, we are 95% confident, that the true average house price lies between $230k and $270k.

Confidence Intervals: Example

Example: Suppose we collect a sample of 50 houses from London. The **sample** mean price is $252,000 with a **sample** standard deviation of $40,000. How good is this estimate of the **true** average house price?

- From a sample of 50 houses, the average price was $252k.
- Using a **95% confidence interval**, we estimate the true average house price is **between $241k and $264k**.



How do we Generate a CI for a Sample Mean?

1. Ensure data follows a normal distribution.
   - If the sample size is large, the CLT tells us the sampling distribution of the mean is approximately normal (**no matter the data distribution**).
2. Compute the **sample** mean, **sample** standard deviation, and **standard error**.
3. Use the **confidence interval formula** for the mean:

$$\mathrm{CI} = \bar{x} \pm z \times \mathrm{SE}$$

Background - Parameters / Statistics

Parameter: a value that characterizes the **population** that we want to study (e.g., mean, standard deviation, regression slope).

Statistic: a single number or measurement that describes a characteristic of a **sample** (e.g. sample mean, sample standard deviation, slope found using OLS).

Statistics are used to **estimate** parameters when the entire population data is unavailable.

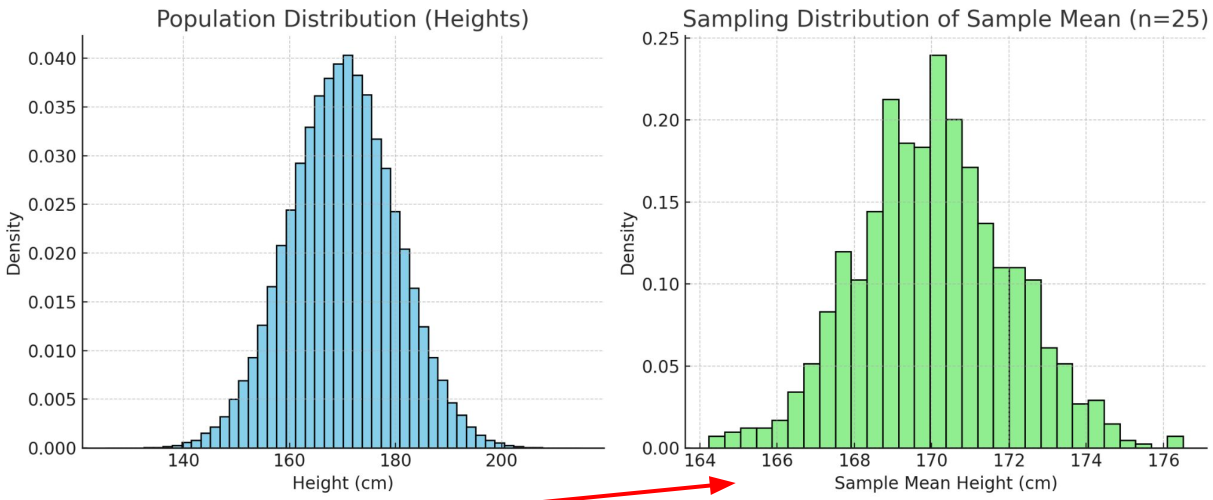
Background - Sampling Distribution

Sampling distribution: the probability distribution of a **statistic** (e.g. mean, variance) that you would get if you repeatedly took samples of the same size from the same population.

Sampling Distribution: Example I

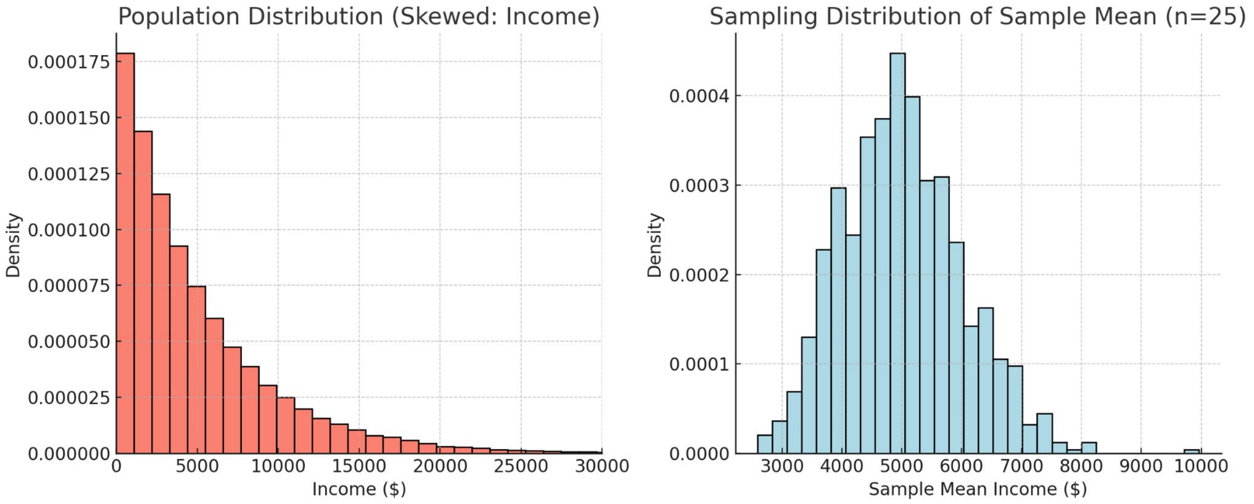
Example of a sampling distribution:

- Suppose the population is all students' heights at a university (μ = 170 cm, σ = 10 cm).
- Take random samples of size n = 25 students over and over again.
- **For each sample**, compute the **sample mean** height.
- If you plot the distribution of those sample means, that's the **sampling distribution** of the **sample mean.**



Sampling Distribution: Example II

- Suppose the population is the annual incomes of students at UWO.
- Take random samples of size n = 25 over and over again.
- For each sample, compute the **sample mean** income.
- If you plot the distribution of those sample means, that's the **sampling distribution** of the sample mean.
- **Even though the population is highly skewed, the sampling distribution is much closer to normal.**



Central Limit Theorem (CLT)

As we just saw, even if we do not know the population distribution, or it is not normal, as long as we have a large enough sample size, we can apply the **Central Limit Theorem.**

CLT $\rightarrow$ for large n, sampling distribution $\approx$ Normal

How large does n have to be?

- This is quite subjective, there's no strict number.
- Common misconception is that it applies for $n > 30$. This is not a hard and fast rule.
- [Helpful Visualization]

Central Limit Theorem (CLT)

When we do an experiment, we don't always know what distribution our data comes from.

- However, using the CLT, we can know that the **sample means** are approximately **normally distributed.**

Reminder: We need a normal distribution to use the CI formula for the mean!

- Required for our z-score.

Confidence Intervals

Confidence Interval (CI): a range of values, calculated from sample data, that will contain the true population parameter in a given proportion of repeated samples (the **confidence level**).

- If we have a 95% confidence interval, this tells us that on average, 95 out of 100 confidence intervals made from sample should include the **real mean.**

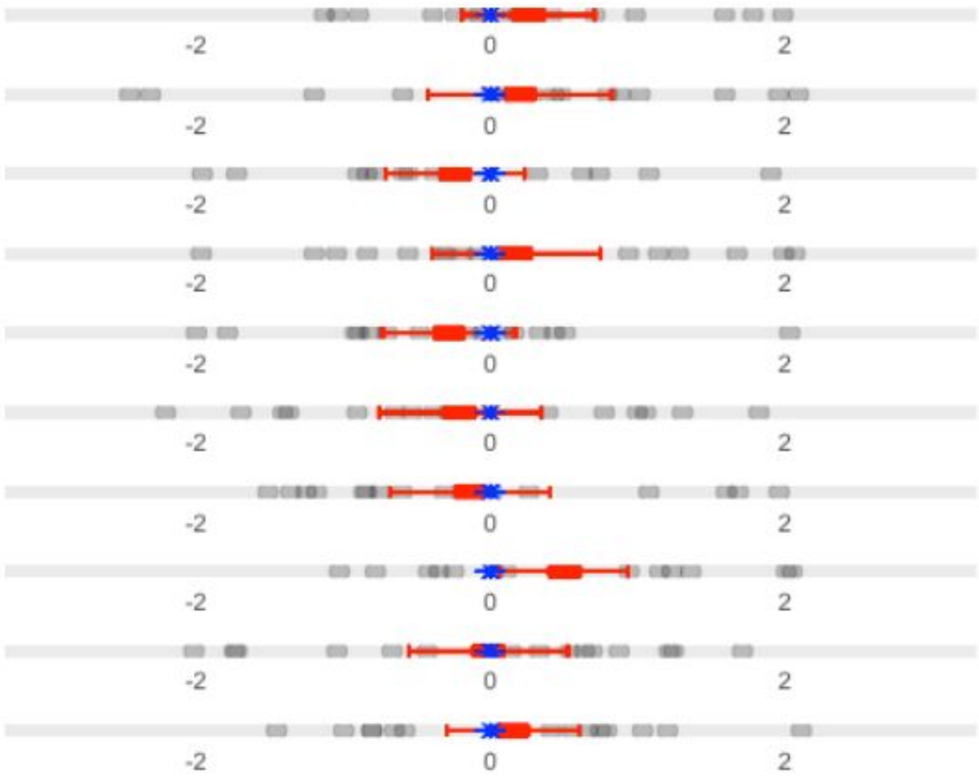
Confidence level: The probability that the confidence interval procedure will capture the true parameter in repeated sampling. We can construct intervals at **different levels** (commonly 90%, 95%, or 99%).

Confidence Intervals

Blue star = population mean

Red interval = CI

Grey dots = samples



If we have a 90% confidence interval, this tells us that on average, 9 out of 10 confidence intervals made from sample should include the **real mean.**

Calculating a CI for a Sample Mean

$$\mathrm{CI} = \bar{x} \pm z \times \mathrm{SE}$$

$$\bar{x} = \frac{\sum x_i}{n}$$

z z-score for the confidence level

$z_{0.90} = 1.645$ for a 90% confidence level

$z_{0.95} = 1.96$ for a 95% confidence level

$z_{0.99} = 2.576$ for a 95% confidence level

$$SE = \frac{s}{\sqrt{n}}$$

SE = Standard Error = Sample standard deviation / square root of sample size

Confidence Intervals: Example

Example: Suppose we have a dataset of volcano eruption times with n = 272 observations. Our *estimate* of the mean of eruption times is 3.49 and standard deviation of 1.14. How good is this estimate (hint: use a 95% confidence interval)?

Step 1) Calculate the standard error

$$\bar{x} = 3.49$$
$$\sigma = 1.14$$
$$n = 272$$

$$SE = \frac{\sigma}{\sqrt{n}}$$
$$= \frac{1.14}{\sqrt{272}}$$
$$\approx 0.07$$

Confidence Intervals: Example

Example: Suppose we have a dataset of volcano eruption times with n = 272 observations. Our *estimate* of the mean of eruption times is 3.49 and standard deviation of 1.14. How good is this estimate (hint: use a 95% confidence interval)?

Step 2) Build the confidence interval

Assuming that our distribution of sample means is normal, we will use the formula for a 95% CI

$$\mathrm{CI} = \bar{x} \pm z_{0.95} \times \mathrm{SE}$$

$\bar{x} = 3.49$ $\quad z_{0.95} = 1.96$ $\quad SE = 0.07$

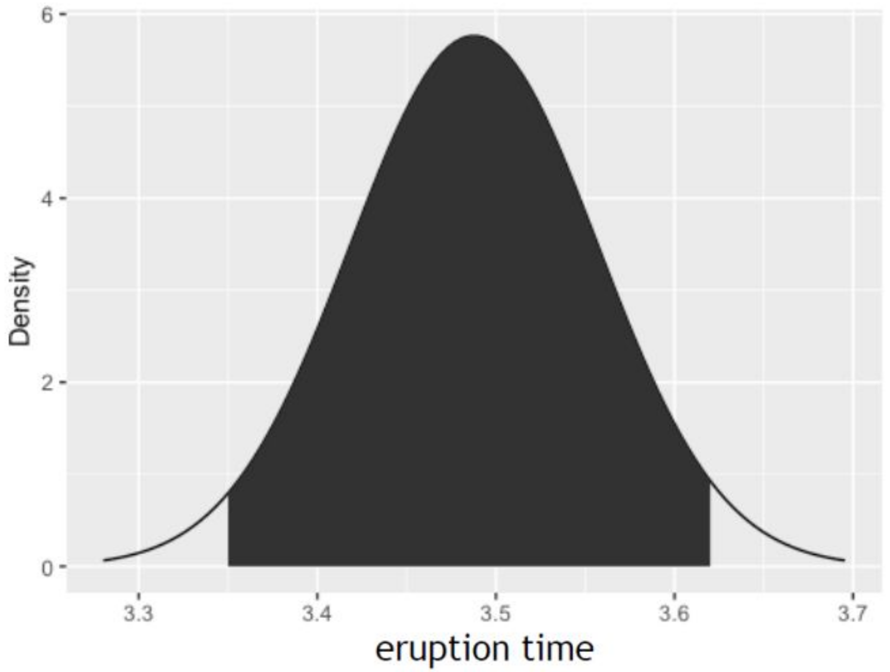
$$\mathrm{CI} = 3.49 \pm 1.96 \times 0.07$$

$$\mathrm{CI} = \{3.35, 3.63\}$$

Confidence Intervals: Example

Example: Suppose we have a dataset of volcano eruption times with n = 272 observations. Our *estimate* of the mean of eruption times is 3.49 and standard deviation of 1.14. How good is this estimate (hint: use a 95% confidence interval)?

$$\mathrm{CI} = \{3.35, 3.63\}$$



Special Case: Bootstrap

There are several scenarios in which we would not want to use the CLT to build a confidence interval:

- **The sample size is small and CLT might not give a good approximation.**
- The underlying population distribution is highly skewed or unknown.
- The statistic of interest is not the mean (e.g., median, regression coefficient, correlation, etc.).

In these cases, we can use **bootstrap** instead.

What is Bootstrap?

Bootstrap: a method of estimating the distribution of an estimator by resampling.

Pros: no distributional assumptions, works with any statistic.

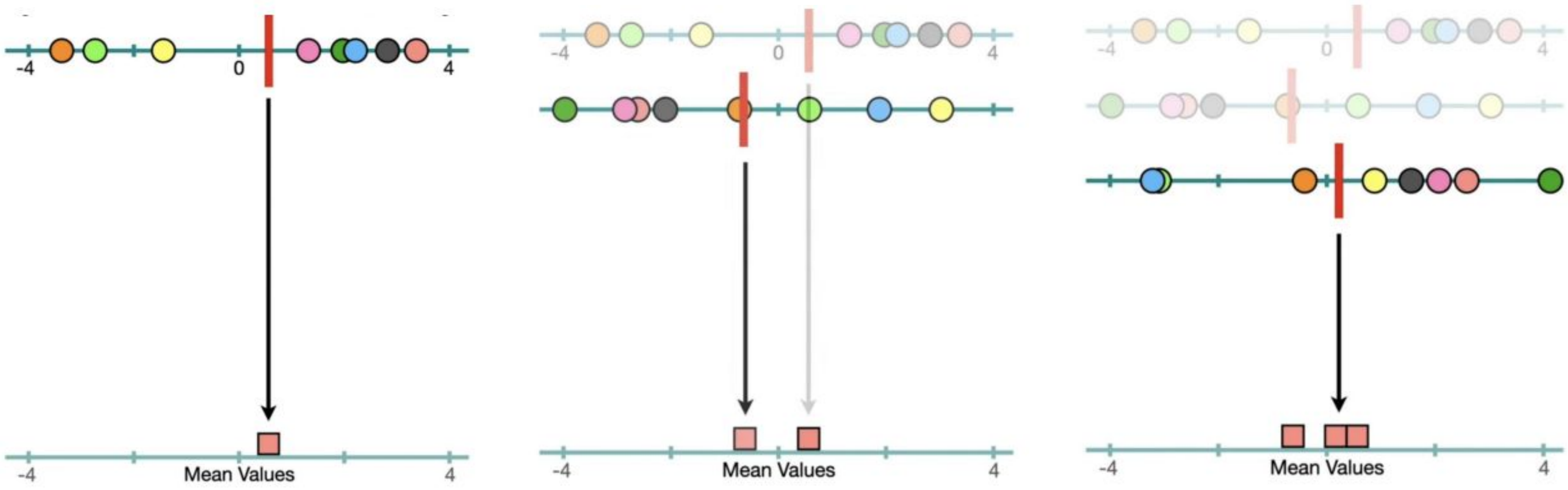
Cons: computationally expensive, can be unreliable with very small samples.

How does Bootstrapping Work?

Bootstrap procedure:

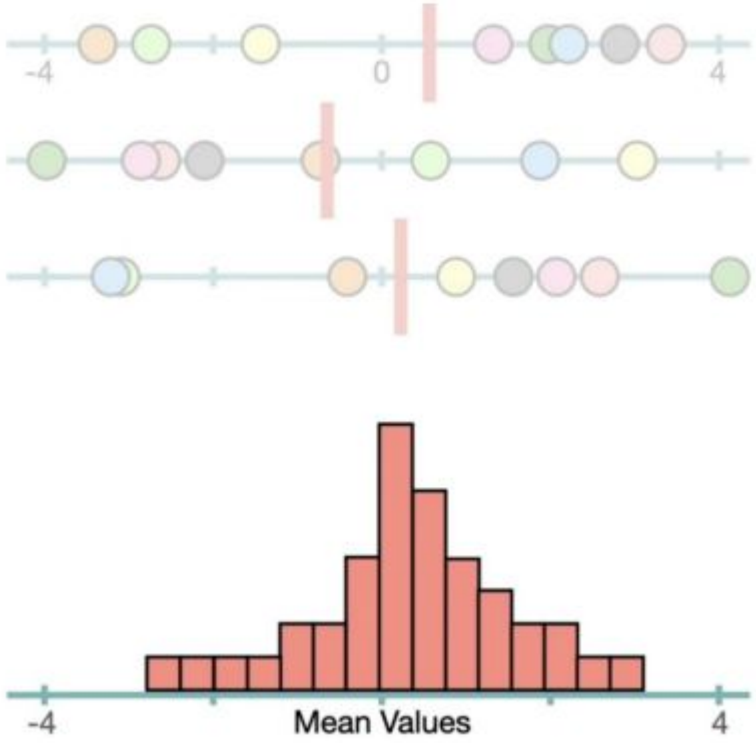
1. Resample *n* data points **with replacement** from sample data.
2. Compute the statistic of interest (e.g. mean, median, regression coefficient) for each new sample.
3. Repeat 1 and 2 *k* times.
   a. *k* is often > 1000
4. Use these **bootstrap statistics** to construct a **confidence interval**.
   a. Take the lower and upper percentiles for your confidence level (e.g., 2.5th and 97.5th percentiles for a 95% CI).

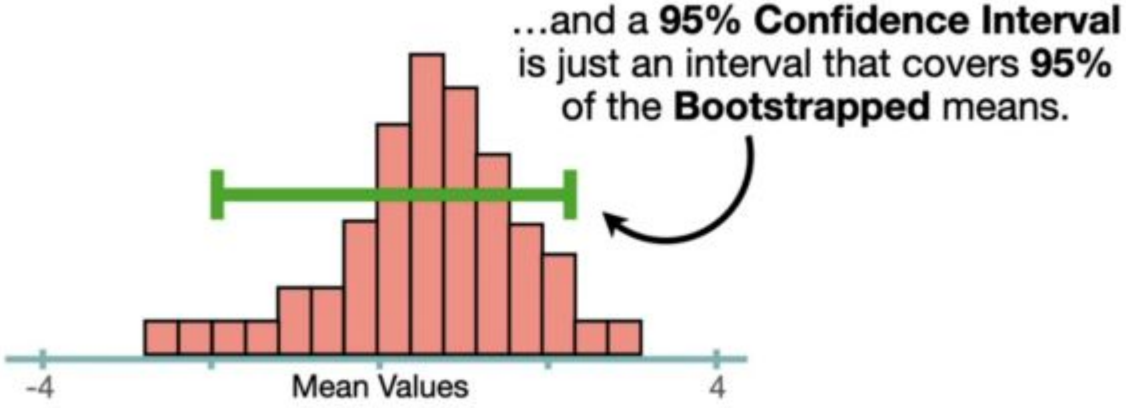
How does Bootstrapping Work?



How does Bootstrapping Work?

After *k* repetitions, we can calculate the construct the CI.





Summary

- We can use **confidence intervals (CIs) to quantify uncertainty.**
- There are 2 main ways to construct CIs:

  1. When you know the distribution (or can assume normality / use CLT), calculate uncertainty **based on specified confidence level.**

  $$\mathrm{CI} = \bar{x} \pm z \times \mathrm{SE}$$

  2. If you do not know distribution or cannot use the **Central Limit Theorem (CLT)**, **bootstrapping** can be used.

  - Repeatedly resample with replacement from observed data. Construct CI by taking lower and upper percentiles for your confidence level.
- The **more data** that we have, the **more confident** we can be in our estimate.